CCN Level - 2 Project Report

Design and Implementation of a VLAN-Segmented Greenhouse IoT Network with Centralized Monitoring and Remote Access

Course: Computer Communication Networks (CCN) — Level 2 Project
Platform: Cisco Packet Tracer

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Class and Section: V Sem E Section

Abstract

The increasing use of Internet of Things (IoT) devices in agriculture enables automated monitoring and control of environmental conditions. However, integrating IoT devices with office networks without proper segmentation can introduce security risks and unnecessary network traffic.

This project presents the design and implementation of a VLAN-segmented greenhouse IoT network using Cisco Packet Tracer. The network consists of a greenhouse IoT network, an office network, and a dedicated server network.

Dynamic Host Configuration Protocol (DHCP) automates IP address allocation for IoT devices. Extended Access Control Lists (ACLs) restrict direct access from the office network to the greenhouse IoT network. Static routing enables communication between the edge router and an ISP router, while Network Address Translation (NAT) and Port Address Translation (PAT) are configured to support remote access through a single public-facing IP address.

An IoT server provides a centralized interface for device registration, monitoring, and control. The implementation demonstrates network segmentation, IP addressing, routing, traffic filtering, and IoT integration. Public-IP-based HTTP access remains subject to further verification in the current implementation.

1. Introduction

The Internet of Things has become an important component of modern agriculture, enabling the monitoring of environmental conditions and the remote operation of equipment.

Greenhouse systems can use sensors to monitor parameters such as temperature, water level, and smoke, while actuators control operations such as irrigation and drainage.

A network that connects these devices must provide reliable communication, centralized management, and appropriate security controls. Placing all devices in a single network may allow unnecessary communication between IoT devices and office computers.

To address these concerns, this project implements a segmented greenhouse network using Cisco Packet Tracer. The network separates IoT devices, office systems, and the central IoT server into distinct VLANs. Routing and access control policies regulate communication between these networks.

The project also incorporates a remote administration network and a public-facing edge router to demonstrate routing and NAT/PAT concepts relevant to computer communication networks.

2. Problem Statement

Problem Statement

Design a VLAN-segmented Cisco Packet Tracer network that connects greenhouse IoT sensors and actuators to a central IoT server, isolates them from office devices, and enables secure remote monitoring and control through a single public IP.

Objectives

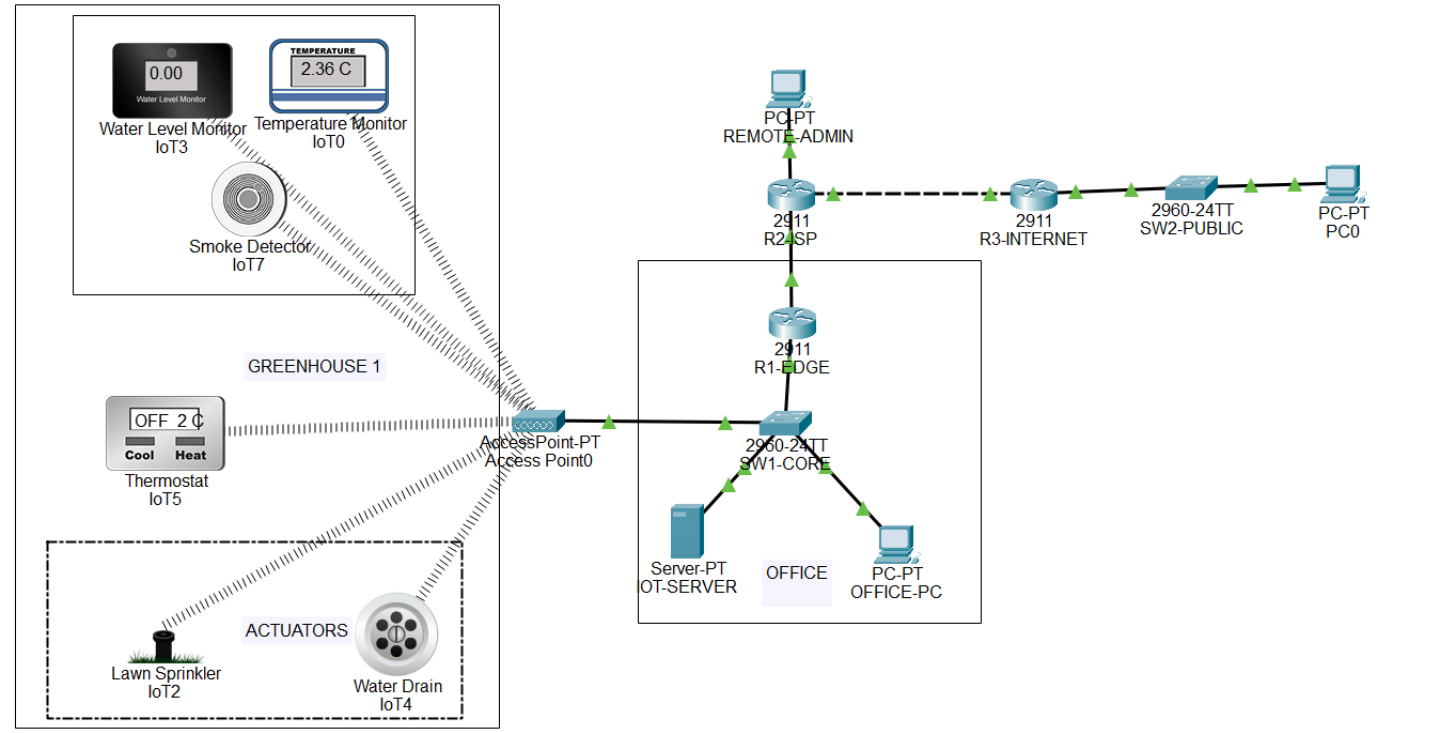
The primary objectives are:

1. Create separate VLANs for greenhouse IoT devices, office devices, and servers.
2. Configure DHCP for automatic IP address allocation to IoT devices.
3. Use extended ACLs to restrict unauthorized communication from the office VLAN to the IoT VLAN.
4. Configure static routing between the edge router and ISP router.
5. Implement NAT/PAT and static port forwarding to demonstrate access through a public IP.
6. Integrate IoT sensors and actuators with a centralized IoT server.
7. Test connectivity, network isolation, routing, and remote access.

3. Network Requirements

Network	Purpose	Network address
VLAN 10 — GREENHOUSE_IOT	Greenhouse sensors and actuators	192.168.10.0/24
VLAN 20 — IOT_SERVERS	Central IoT server	192.168.20.0/24
VLAN 30 — OFFICE	Office workstation	192.168.30.0/24
ISP-facing link	Connection between edge and ISP routers	203.0.113.0/30
Remote administration network	Remote administrator and ISP router	198.51.100.0/24

4. Network Topology Design



The network uses a hierarchical logical arrangement consisting of an edge router, a core switch, a central server, and end devices.

Device	Role
R1-EDGE — Cisco 2911	Inter-VLAN routing, ACL enforcement, NAT/PAT, and connectivity to the ISP
R2-ISP — Cisco 2911	Simulated ISP router and remote-network connectivity
R3-INTERNET – Cisco 2911	Represents an intermediate internet router that connects the ISP network to the public network.
SW1-CORE — Cisco 2960	VLAN creation, access-port assignment, and Layer 2 switching
IOT-SERVER — Server-PT	IoT device registration and centralized management
OFFICE-PC	Represents an office workstation
REMOTE-ADMIN	Represents an administrator accessing the network remotely
PC0 – PC-PT	Represents an external public-network computer used to test connectivity from outside the internal network.
AccessPoint0	Provides wireless connectivity for greenhouse IoT devices
IoT sensors and actuators	Monitor conditions and perform greenhouse-related operations

The router-on-a-stick approach allows multiple VLANs to use one physical router interface through logical sub-interfaces. This reduces the number of physical interfaces required while supporting routing between the three internal networks.

The core switch provides Layer 2 segmentation, while the edge router acts as the controlled boundary between internal networks and the simulated ISP. A separate server VLAN provides a dedicated network for the centralized IoT service.

5. IP Addressing and Subnetting

IPv4 addressing is used throughout the network. Each internal VLAN is assigned a separate /24 subnet, while the point-to-point connection between the edge and ISP routers uses a /30 subnet.

Device/interface	IP address	Subnet mask	Default gateway / next hop
R1-EDGE G0/0.10	192.168.10.1	255.255.255.0	—
R1-EDGE G0/0.20	192.168.20.1	255.255.255.0	—
R1-EDGE G0/0.30	192.168.30.1	255.255.255.0	—
R1-EDGE G0/1	203.0.113.2	255.255.255.252	—
R2-ISP G0/0	203.0.113.1	255.255.255.252	—
R2-ISP G0/1	198.51.100.1	255.255.255.0	—
R3-INTERNET G0/0	198.51.100.2	255.255.255.0	—
R3-INTERNET G0/1	203.0.113.1	255.255.255.252	—
IOT-SERVER	192.168.20.10	255.255.255.0	192.168.20.1
OFFICE-PC	192.168.30.10	255.255.255.0	192.168.30.1
REMOTE-ADMIN	198.51.100.10	255.255.255.0	198.51.100.1
PC0 – PC-PT	203.0.113.10	255.255.255.0	203.0.113.1
Greenhouse IoT devices	DHCP-assigned addresses in 192.168.10.0/24	255.255.255.0	192.168.10.1

A /24 subnet uses a subnet mask of 255.255.255.0 and provides 254 conventional usable host addresses. Separate /24 networks are used for the IoT, server, and office VLANs.

The router-to-router link uses a /30 subnet with the addresses 203.0.113.1 and 203.0.113.2 assigned to the two router interfaces.

The address ranges 203.0.113.0/24 and 198.51.100.0/24 are reserved for documentation. Their use here represents a simulated public-facing network in Packet Tracer and does not provide Internet reachability in a real deployment.

6. Protocols and Technologies Used

VLANs

VLANs logically separate devices into different broadcast domains.

- VLAN 10: GREENHOUSE_IOT
- VLAN 20: IOT_SERVERS
- VLAN 30: OFFICE

This separation supports network organization and provides a basis for enforcing communication policies.

Inter-VLAN Routing

Router-on-a-stick routing is implemented using sub-interfaces on R1-EDGE. Each sub-interface is associated with a VLAN through 802.1Q encapsulation and configured with that VLAN's default gateway.

IEEE 802.1Q Trunking

IEEE 802.1Q tagging allows multiple VLANs to traverse a single physical Ethernet link. The router-facing switch port must be configured as a trunk, with the appropriate VLANs permitted.

The access ports connected to end devices are assigned to their respective VLANs.

DHCP

DHCP provides automatic address allocation for greenhouse IoT devices. An address range is excluded for infrastructure addresses, while the remaining addresses are allocated to clients.

Extended ACLs

An extended ACL restricts direct IP communication from the office subnet to the greenhouse IoT subnet while permitting other traffic not explicitly denied.

Static Routing

Static routes are configured to provide connectivity between the internal networks and the simulated ISP.

NAT and PAT

NAT translates private addresses for communication through the edge router. PAT allows multiple internal connections to share the address of the outside interface by using different transport-layer port numbers.

Wireless Networking

An access point provides wireless connectivity to greenhouse IoT devices. The devices must use the configured SSID and compatible security credentials to join the wireless network.

WPA2-PSK is used in the simulated wireless configuration.

IoT Services

The central server provides an interface for registering IoT devices and viewing their status. Sensors and actuators are registered with the server, enabling centralized monitoring and supported device control through the dashboard.

7. Implementation and Configuration

The implementation was carried out in the following stages.

Step 1: Create the Topology

The routers, core switch, server, office workstation, remote administrator, wireless access point, sensors, and actuators were placed in Cisco Packet Tracer and connected according to the network design.

Step 2: Configure VLANs

VLANs 10, 20, and 30 were created on the core switch and assigned to the appropriate access ports.

Step 3: Configure Trunking

The switch port connected to R1-EDGE was configured to carry the required VLANs using IEEE 802.1Q trunking. Trunk operation should be verified using the switch's operational status commands.

Step 4: Configure Inter-VLAN Routing

Router sub-interfaces were created and assigned the corresponding VLAN IDs and gateway addresses.

Step 5: Configure IP Addresses and DHCP

Static addressing was configured for the server and office workstation. DHCP was configured on R1-EDGE to allocate addresses to the greenhouse IoT devices.

Step 6: Apply the Extended ACL

The office-to-IoT ACL was configured and applied inbound on the office sub-interface. Connectivity tests were then performed to confirm that direct office-to-IoT communication was blocked while access to the server remained possible.

Step 7: Configure the ISP Connection

The R1-EDGE and R2-ISP interfaces were configured using the 203.0.113.0/30 link. The remote administration network was assigned the 198.51.100.0/24 subnet.

Step 8: Configure NAT/PAT

PAT and static TCP port forwarding were configured on R1-EDGE. NAT translations were inspected while remote HTTP requests were attempted.

Step 9: Configure the IoT Server

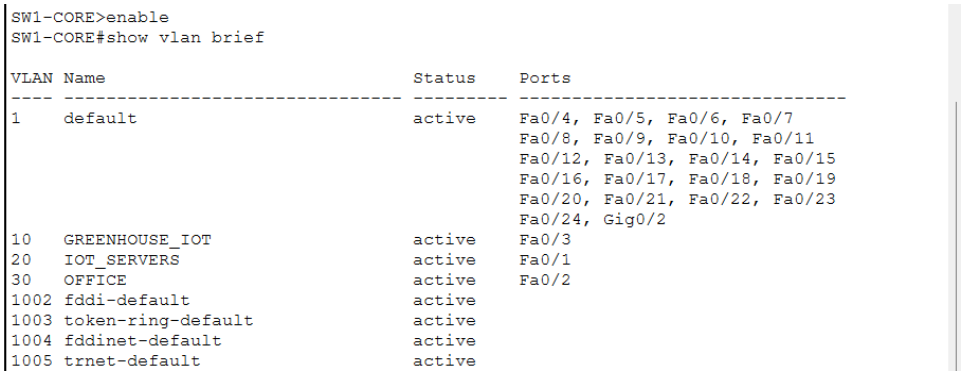
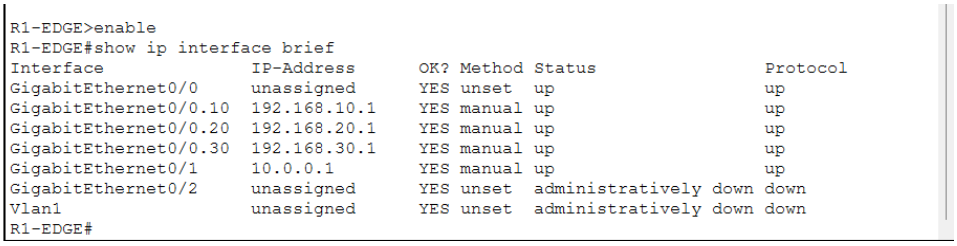
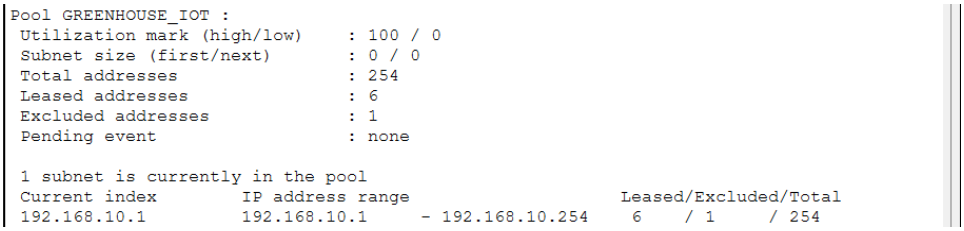
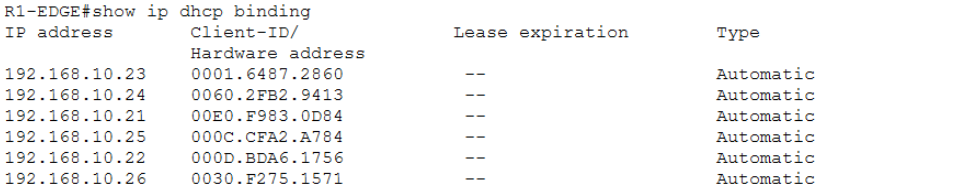
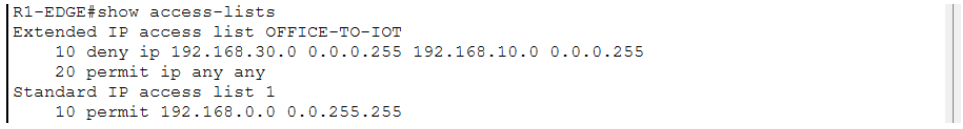
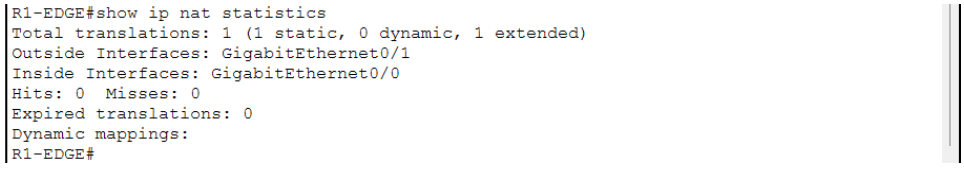
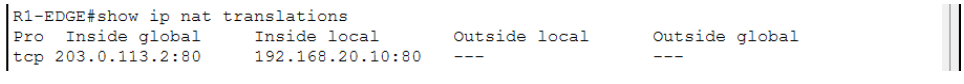
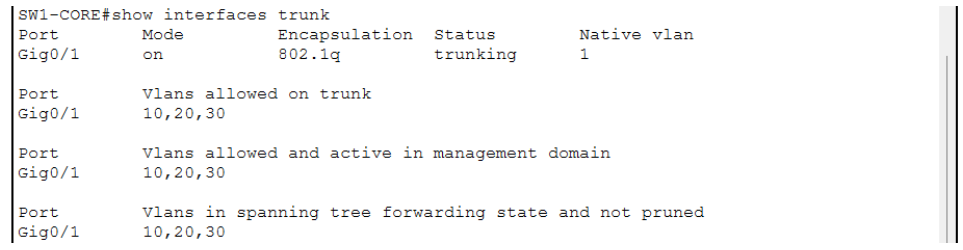
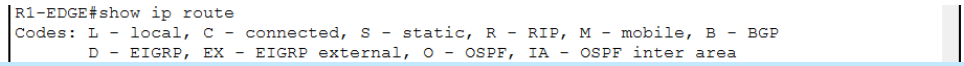
HTTP and IoT services were enabled. IoT devices were connected to the wireless network and registered with the central server.

Step 10: Verify Remote Administration

The remote administrator successfully accessed the IoT registration page using the private server address. Public-IP HTTP access was then tested separately.

8. Testing and Verification

The following commands can be used to capture supporting evidence:

Command	Figure (results)
show vlan brief	
show ip interface brief	
show ip dhcp pool	
show ip dhcp binding	
show access-lists	
show ip nat statistics	
show ip nat translations	
show interfaces trunk	
	

show ip route

```
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is 10.0.0.2 to network 0.0.0.0

     10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       10.0.0.0/30 is directly connected, GigabitEthernet0/1
L       10.0.0.1/32 is directly connected, GigabitEthernet0/1
     192.168.10.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.10.0/24 is directly connected, GigabitEthernet0/0.10
L       192.168.10.1/32 is directly connected, GigabitEthernet0/0.10
     192.168.20.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.20.0/24 is directly connected, GigabitEthernet0/0.20
L       192.168.20.1/32 is directly connected, GigabitEthernet0/0.20
     192.168.30.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.30.0/24 is directly connected, GigabitEthernet0/0.30
L       192.168.30.1/32 is directly connected, GigabitEthernet0/0.30
S*   0.0.0.0/0 [1/0] via 10.0.0.2
```

```
R2-ISP>show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       10.0.0.0/30 is directly connected, GigabitEthernet0/0
L       10.0.0.2/32 is directly connected, GigabitEthernet0/0
     192.168.0.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.0.0/24 is directly connected, GigabitEthernet0/1
L       192.168.0.1/32 is directly connected, GigabitEthernet0/1
S    192.168.10.0/24 [1/0] via 10.0.0.1
S    192.168.20.0/24 [1/0] via 10.0.0.1
S    192.168.30.0/24 [1/0] via 10.0.0.1
```

show interfaces status

```
SW1-CORE>enable
SW1-CORE#show interfaces status
Port      Name               Status       Vlan       Duplex  Speed Type
Fa0/1                        connected    20         a-full  a-100 10/100BaseTX
Fa0/2                        connected    30         a-full  a-100 10/100BaseTX
Fa0/3                        connected    10         a-full  a-100 10/100BaseTX
Fa0/4                        notconnect   1          auto    auto  10/100BaseTX
Fa0/5                        notconnect   1          auto    auto  10/100BaseTX
Fa0/6                        notconnect   1          auto    auto  10/100BaseTX
Fa0/7                        notconnect   1          auto    auto  10/100BaseTX
Fa0/8                        notconnect   1          auto    auto  10/100BaseTX
Fa0/9                        notconnect   1          auto    auto  10/100BaseTX
Fa0/10                       notconnect   1          auto    auto  10/100BaseTX
Fa0/11                       notconnect   1          auto    auto  10/100BaseTX
Fa0/12                       notconnect   1          auto    auto  10/100BaseTX
Fa0/13                       notconnect   1          auto    auto  10/100BaseTX
Fa0/14                       notconnect   1          auto    auto  10/100BaseTX
Fa0/15                       notconnect   1          auto    auto  10/100BaseTX
Fa0/16                       notconnect   1          auto    auto  10/100BaseTX
Fa0/17                       notconnect   1          auto    auto  10/100BaseTX
Fa0/18                       notconnect   1          auto    auto  10/100BaseTX
Fa0/19                       notconnect   1          auto    auto  10/100BaseTX
Fa0/20                       notconnect   1          auto    auto  10/100BaseTX
Fa0/21                       notconnect   1          auto    auto  10/100BaseTX
Fa0/22                       notconnect   1          auto    auto  10/100BaseTX
Fa0/23                       notconnect   1          auto    auto  10/100BaseTX
Fa0/24                       notconnect   1          auto    auto  10/100BaseTX
Gig0/1                       connected    trunk      a-full  a-100 10/100/1000BaseTX
Gig0/2                       notconnect   1          auto    auto  10/100/1000BaseTX
```

show mac address-table

```
SW1-CORE#show mac address-table
          Mac Address Table
-------------------------------------------

Vlan    Mac Address       Type        Ports
----    -----------       --------    -----

   1    00e0.8f40.6a01    DYNAMIC     Gig0/1
  10    0001.6487.2860    DYNAMIC     Fa0/3
  10    000c.cfa2.a784    DYNAMIC     Fa0/3
  10    000d.bda6.1756    DYNAMIC     Fa0/3
  10    0030.f275.1571    DYNAMIC     Fa0/3
  10    0060.2fb2.9413    DYNAMIC     Fa0/3
  10    00e0.8f40.6a01    DYNAMIC     Gig0/1
  10    00e0.f983.0d84    DYNAMIC     Fa0/3
  20    0003.e433.e832    DYNAMIC     Fa0/1
  20    00e0.8f40.6a01    DYNAMIC     Gig0/1
  30    00e0.8f40.6a01    DYNAMIC     Gig0/1
```

Test	Procedure	Result
VLAN configuration	Execute show vlan brief on SW1-CORE	Successful
802.1Q trunking	Execute show interfaces trunk	Verified
Inter-VLAN routing	Ping the IoT server from OFFICE-PC	Successful
DHCP allocation	Execute show ip dhcp pool and show ip dhcp binding	Successful
ACL enforcement	Attempt communication from OFFICE-PC to an IoT device	Blocked as intended
Static routing	Inspect routing tables on R1-EDGE and R2-ISP	Verified
Remote connectivity	Ping the public-facing router from REMOTE-ADMIN	Successful

9. Results and Discussion

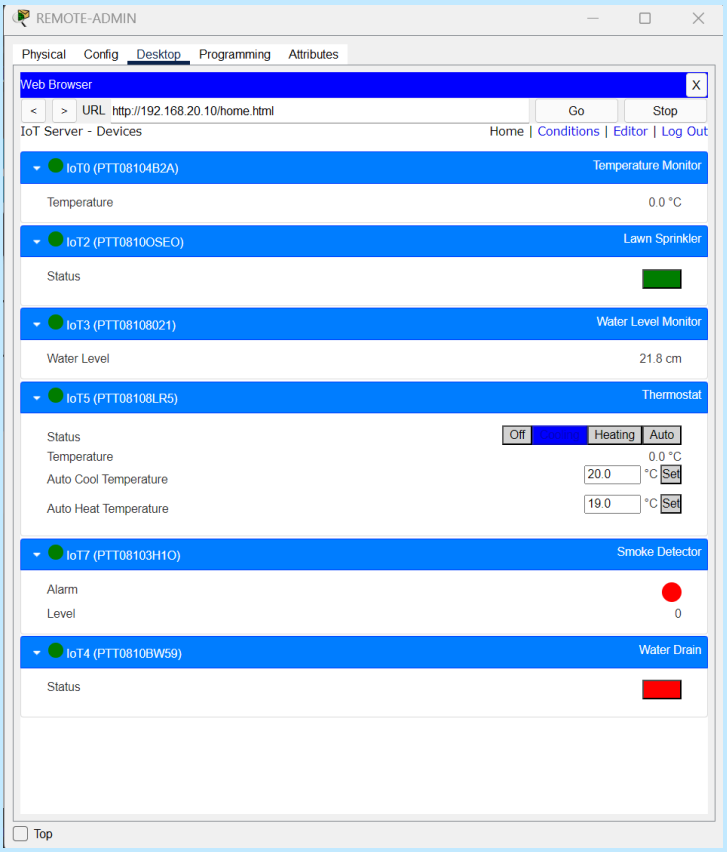
The implemented network successfully demonstrates the segmentation and centralized management of greenhouse IoT devices using Cisco Packet Tracer.

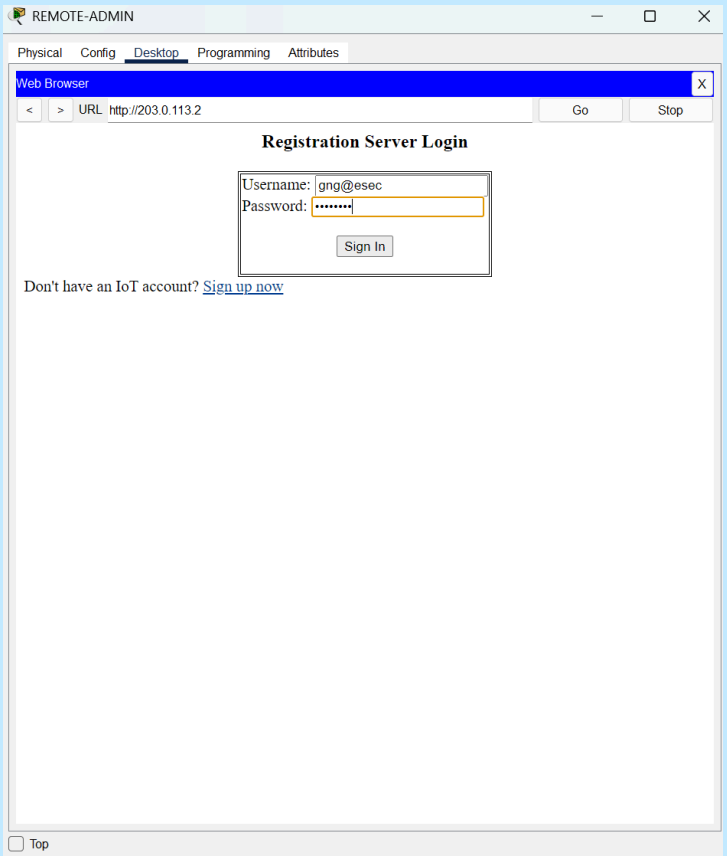
The three VLANs provide logical separation between greenhouse IoT devices, office workstations, and the central server. Inter-VLAN routing enables the required communication between these networks, while extended ACLs restrict direct access from the office network to the greenhouse IoT network.

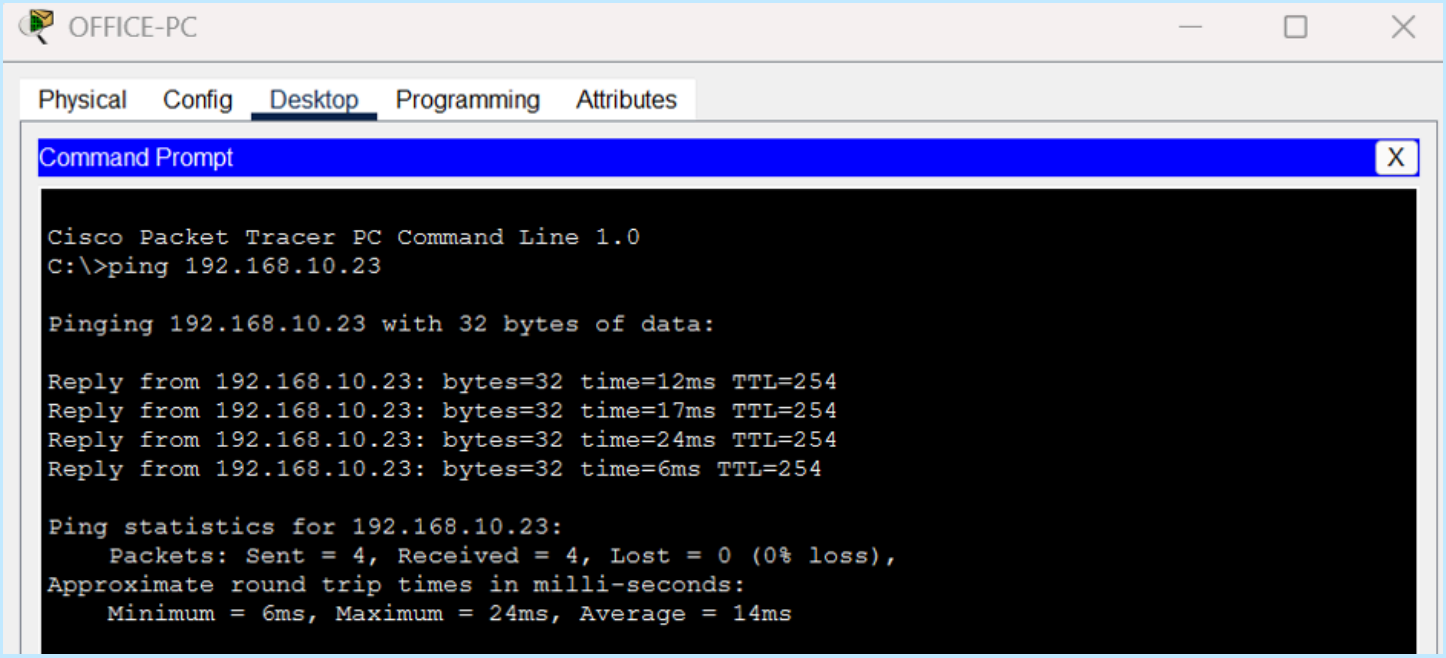
DHCP successfully assigns IP addresses to wireless IoT devices, reducing the need for manual configuration. Static routing provides connectivity between the internal network and the simulated ISP network.

NAT/PAT enables address translation at the edge router, while static TCP port forwarding allows the remote administrator to access the central IoT server through a single public-facing IP address. The IoT server successfully supports device registration and centralized dashboard access. The sprinkler and water drain operations were also tested through the dashboard, demonstrating basic remote actuator control and water-level monitoring. The successful public-IP access test confirms that the remote administration requirement has been demonstrated within the simulated environment.

Overall, the implementation achieves the primary networking and IoT integration objectives defined in the project problem statement.


REMOTE-ADMIN
Physical
Config
Desktop
Programming
Attributes
Web Browser
X
<
>
URL
http://192.168.20.10/home.html
Go
Stop
IoT Server - Devices
Home | Conditions | Editor | Log Out
IoT0 (PTT08104B2A)
Temperature Monitor
Temperature
0.0 °C
IoT2 (PTT0810OSEO)
Lawn Sprinkler
Status
IoT3 (PTT08108021)
Water Level Monitor
Water Level
21.8 cm
IoT5 (PTT08108LR5)
Thermostat
Status
Off
Heating
Auto
Temperature
0.0 °C
Auto Cool Temperature
20.0
°C
Set
Auto Heat Temperature
19.0
°C
Set
IoT7 (PTT08103H1O)
Smoke Detector
Alarm
Level
0
IoT4 (PTT0810BW59)
Water Drain
Status
Top


REMOTE-ADMIN
Physical
Config
Desktop
Programming
Attributes
Web Browser
X
<
>
URL
http://203.0.113.2
Go
Stop
Registration Server Login
Username:
gng@esec
Password:
Sign In
Don't have an IoT account? Sign up now
Top



10. Challenges and Troubleshooting

IoT Device Connectivity

Some IoT devices initially had incorrect wireless settings and failed to obtain valid network addresses.

Resolution: The wireless SSID, security mode, and passphrase were aligned with the access point configuration. Successful DHCP allocation and server registration were then verified.

Office-to-IoT Isolation

The office network needed to be prevented from directly reaching greenhouse IoT devices without unnecessarily blocking access to the central server.

Resolution: An extended ACL was configured on the office VLAN's router subinterface. Testing confirmed that direct office-to-IoT traffic was blocked while office-to-server connectivity remained available.

Public-IP HTTP Timeout

The remote administrator could access the registration page through the private server address but received a timeout when accessing the public-facing address.

Troubleshooting performed: Router interfaces, routing, static NAT configuration, NAT translations, and basic reachability were inspected. The translation table showed entries associated with the attempted public HTTP connection.

11. Limitations and Future Enhancements

Although the project successfully demonstrates VLAN segmentation, routing, access control, IoT integration, and public-IP-based remote access, certain enhancements can be considered for future development.

1. **HTTPS implementation:** Introduce encrypted communication for remote dashboard access, provided the simulated server supports HTTPS.
2. **Enhanced access control:** Restrict access to exposed services using more specific ACL rules and trusted remote sources.
3. **Switch port security:** Configure MAC-address-based port security to prevent unauthorized devices from connecting to selected switch ports.
4. **Network redundancy:** Introduce redundant links and configure Spanning Tree Protocol to improve resilience.
5. **Advanced IoT automation:** Extend sensor-based automation for temperature monitoring, smoke detection, irrigation, and water management.
6. **Dynamic routing:** Explore OSPF in a larger network topology containing multiple routers.

These enhancements could improve the security, scalability, and reliability of the proposed greenhouse network.

12. Conclusion

This project successfully demonstrates the design and implementation of a VLAN-segmented greenhouse IoT network using Cisco Packet Tracer.

The network integrates greenhouse sensors and actuators with a centralized IoT server while maintaining logical separation between the IoT, office, and server networks.

The implementation incorporates IEEE 802.1Q trunking, inter-VLAN routing, DHCP, IPv4 subnetting, extended ACLs, static routing, and NAT/PAT. These technologies provide structured IP addressing, controlled communication, and connectivity between the internal networks and the simulated remote administration network.

The successful implementation of static NAT port forwarding enables the remote administrator to access the IoT server through a single public-facing IP address. The registered greenhouse devices can be monitored and controlled through the centralized dashboard.

The project provided practical exposure to network segmentation, routing, address translation, access control, wireless connectivity, and IoT integration. It also demonstrated the importance of systematic troubleshooting and configuration verification in developing reliable network solutions.

Thus, the project achieves its primary objective of demonstrating a segmented greenhouse IoT network with centralized management and public-IP-based remote access within the Cisco Packet Tracer simulation environment.

13. References

1. Cisco Systems, *Cisco Packet Tracer* — network simulation and device configuration environment.
2. Cisco Systems, *Cisco IOS IP Addressing Services Configuration Guide* — DHCP and NAT configuration concepts.
3. Cisco Systems, *Cisco IOS Security Configuration Guide* — access control lists and network security.
4. Cisco Systems, *VLAN and IEEE 802.1Q Trunking Configuration Documentation*.
5. Cisco Systems, *Routing Configuration Documentation* — static routing and inter-network connectivity.